

Managing Pluggable Authentication Modules (PAM)



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Overview



PAM

- What and why PAM
- Creating home directories on user login
- Locking user accounts
 - ordering correctly in PAM
 - pam_tally2



PAM allows modules to be loaded for authentication services, so features can be shared



```
$ man pam.d
```

Documentation

To view the makeup of a PAM file we can open the pam.d manual page

```
$ man pam_mkhomedir
```

Help on Module

For individual modules, we can investigate the manual page for the specific module

PAM Types

Auth

Modules execute during authentication

Account

Modules execute after authentication checking for account restrictions, such as time of day, or active sessions

Password

Modules execute during a change of authentication token such as a password

Session

Modules execute before a service is presented to a user but after other restrictions have been passed. Home directory creation is an example



PAM Control

Required

Modules must return success, but other modules are run no matter

Requisite

As above but other modules are not executed if this fails

Sufficient

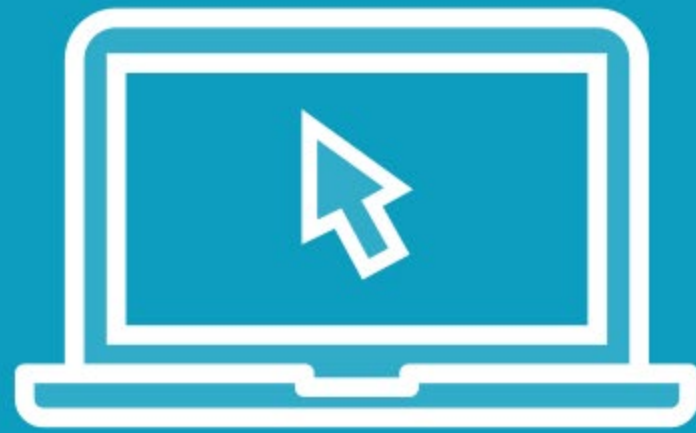
If the module returns success, then no other module needs to be executed

Optional

The success or failure of this module is not important unless it is the only module executed



Demo



Investigating PAM

- List PAM configuration
- Display help on modules and configuration



Having PAM create home directories may be useful where central user accounts are used, and the home directory is not created with the user



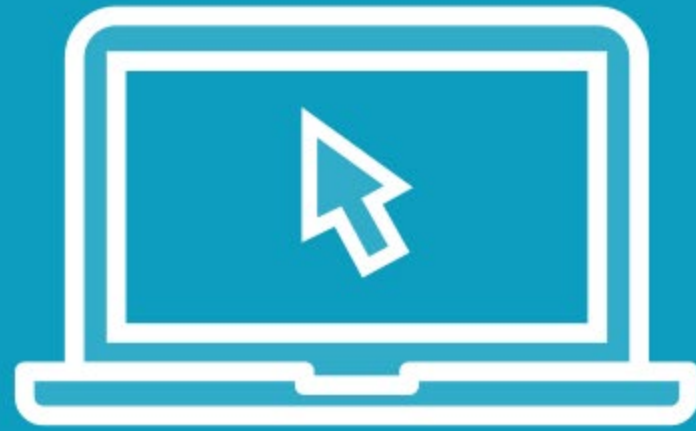
```
$ echo "session optional pam_mkhomedir.so" | sudo tee -a /etc/pam.d/common-session
$ su - gandalf

$ exit
```

Configure PAM

Adding the common-session file, the entry is available to all services

Demo



Creating User Home Directories:

- Create user without home directory
- Test access
- Modify PAM



Adding to your security, we can
lock user accounts where a
failed login attempt has
occurred a given number of
times



```
$ echo "auth required pam_tally2.so onerr=fail deny=6 unlock_time=300" | sudo tee -a /etc/pam.d/common-auth
$ su - gandalf
$ sudo pam_tally2 -u gandalf
$ sudo sed -i '$d' /etc/pam.d/common-auth
```

Ordering Incorrect

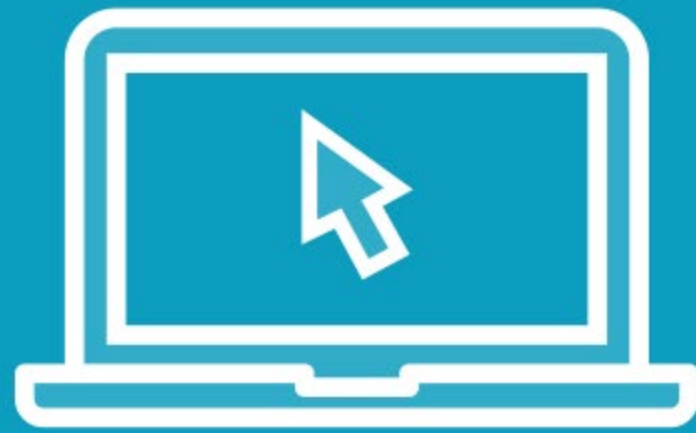
Adding to the end of the file may not work in all cases. Where a wrong password is supplied the tally module may never run if a requisite control happens first


```
$ sudo sed -i "1i auth required pam_tally2.so onerr=fail deny=6 unlock_time=300"  
/etc/pam.d/common-auth  
$ su - gandalf  
$ sudo pam_tally2 -u gandalf
```

Ordering Correct

Having the tally module set early on the bad login is registered before access is denied

Demo



Locking Accounts:

- Appending to the common-auth file
- Viewing failed login count
- Adding the entry in earlier
- Resetting login count



Summary



Working with PAM

- `man pam.d`
- `man pam_mkhomedir`
- `man pam_tally2`
- `/etc/pam.d/common-session`
- `/etc/pam.d/common-auth`
- failures and correct ordering
- `pam_tally2`



Congratulations!

